

# Unit 10, Lesson 1: Creating Box Plots

## Benchmarks

MA.6.DP.1.3 Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile and the maximum. Use this summary of the data to describe the spread and distribution of the data.

MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts.

## Learning Targets

- ☐ I can find key values given a box plot within a real-world context.
- ☐ I can create a box plot to represent data.

Ten sixth-grade students were asked how much sleep, in hours, they usually get on a school night. Their responses are shown.

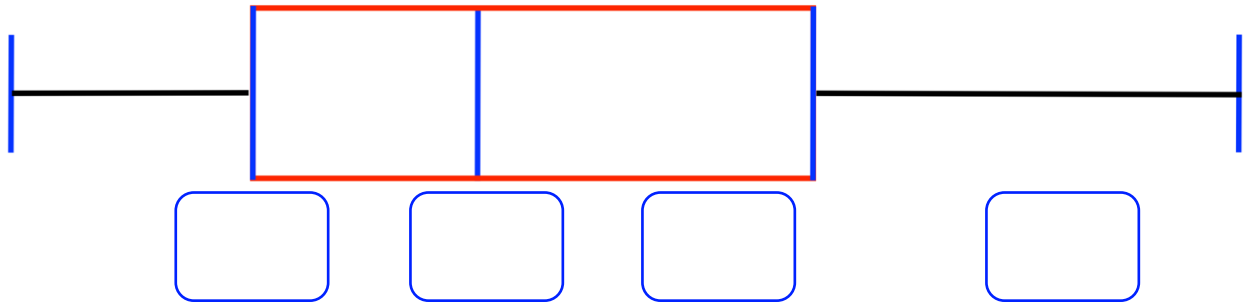
1. Calculate the median number of hours of sleep for the students.

2. Calculate the range for number of hours of sleep for the students.

### 10.1.2 Refresher: Box Plots

A blank box plot is shown. Label the blue boxes below using the words found in the bank.

|                          |                          |               |
|--------------------------|--------------------------|---------------|
| median                   | first quartile ( $Q_1$ ) | minimum (min) |
| third quartile ( $Q_3$ ) | maximum (max)            |               |



1. Complete the statements.

The \_\_\_\_\_ is a measure of spread determined by finding the distance between the first quartile and the third quartile. It measures the spread of the middle \_\_\_\_\_ of the data.

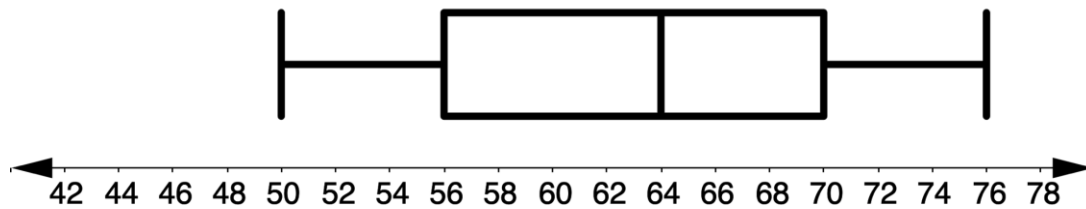
The \_\_\_\_\_ is another measure of spread determined by finding the difference between the largest data value and the smallest data value.

2. Complete each statement.

- \_\_\_\_\_ % of the data falls between the minimum and the first quartile ( $Q_1$ ).
- \_\_\_\_\_ % of the data falls between the minimum and the third quartile ( $Q_3$ ).
- \_\_\_\_\_ % of the data falls between the first quartile ( $Q_1$ ) and the third quartile ( $Q_3$ ).

3. The box plot shows the distribution of the daily number of people who were at a large community pool in July 2018.

**Number of Patrons at Community Pool**



- What is the median number of people at the pool each day?
- What is the maximum number of people?
- What is the minimum number of people?
- Determine the value of the first quartile ( $Q_1$ ) from the box plot.
- Determine the value of the third quartile ( $Q_3$ ) from the box plot.



The **five-number summary** are the values of the minimum, first quartile ( $Q_1$ ), median, third quartile ( $Q_3$ ), and the maximum. These are the values used to create a box plot to represent a data set.

### 10.1.3 Guided Instruction: The Five-Number Summary

1. The ages of 20 customers who attended a special event at a local restaurant are shown in order from least to greatest.

7, 8, 9, 10, 10, 11, 12, 15, 16, 20, 20,  
22, 23, 24, 28, 30, 33, 35, 38, 42

- a. Find the five-number summary of the data set.

Minimum: \_\_\_\_\_

First quartile: \_\_\_\_\_

Median: \_\_\_\_\_

Third quartile: \_\_\_\_\_

Maximum: \_\_\_\_\_

- b. Discuss with your partner if the data the restaurant collected is reflective of the ages of customers who ate at the restaurant.



**Truthful data** is data that is random, representative of the population, and has a large sample size.

2. The birth weights, in ounces, of golden retriever puppies born at a kennel were recorded for the past month. The data is recorded in the table.

|    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|
| 13 | 14 | 15 | 15 | 16 | 16 | 16 | 16 | 17 |
| 17 | 17 | 17 | 17 | 17 | 17 | 18 | 18 | 18 |
| 18 | 18 | 18 | 18 | 18 | 19 | 20 |    |    |



- a. Find the five-number summary of the data set.

Minimum: \_\_\_\_\_

First quartile: \_\_\_\_\_

Median: \_\_\_\_\_

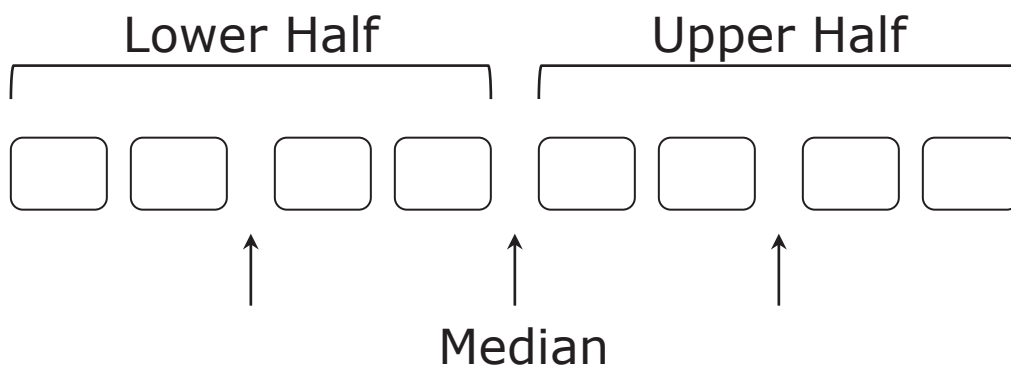
Third quartile: \_\_\_\_\_

Maximum: \_\_\_\_\_

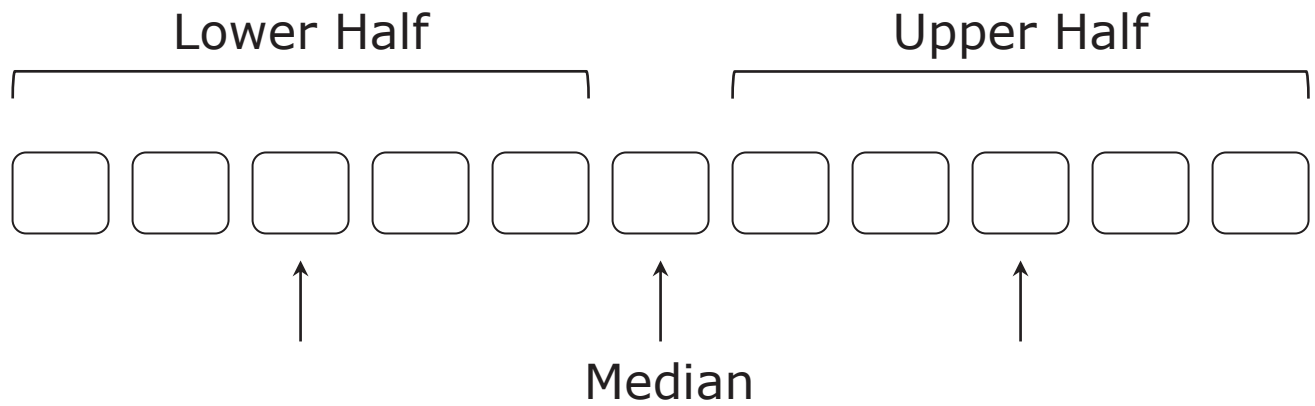
- b. Discuss with your partner why the data may not be a true representation of the birth weight of all puppies.

3. What differences did you notice when calculating the five-number summary for the two different data sets?

An illustration of the five-number summary for an even data set is shown.



An illustration of the five-number summary for an odd data set is shown.



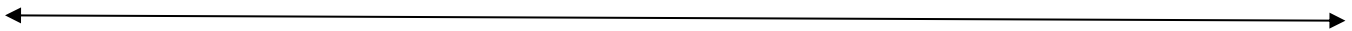
4. Discuss with your partner your observations and any questions you have about the illustrations.

### 10.1.4 Guided Practice: Creating a Box Plot

- Using the data and five-number summary on the birth weight, in ounces, of the golden retriever puppies born at a kennel in the last month, create a box plot from the steps listed.

|    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|
| 13 | 14 | 15 | 15 | 16 | 16 | 16 | 16 | 17 |
| 17 | 17 | 17 | 17 | 17 | 17 | 18 | 18 | 18 |
| 18 | 18 | 18 | 18 | 18 | 19 | 20 |    |    |

Step 1: Label the axis by starting at a number slightly less than the minimum. Increase by the same interval, making sure each interval is equally spaced, until you reach (or pass) the maximum.



Step 2: Place a dot above the axis for each value of the five-number summary.

Step 3: Draw the box from  $Q_1$  to  $Q_3$  with a line inside the box to represent the median.

Step 4: Connect the minimum and the maximum to each side of the box with a line.

Step 5: Include a title and a label for the axis.

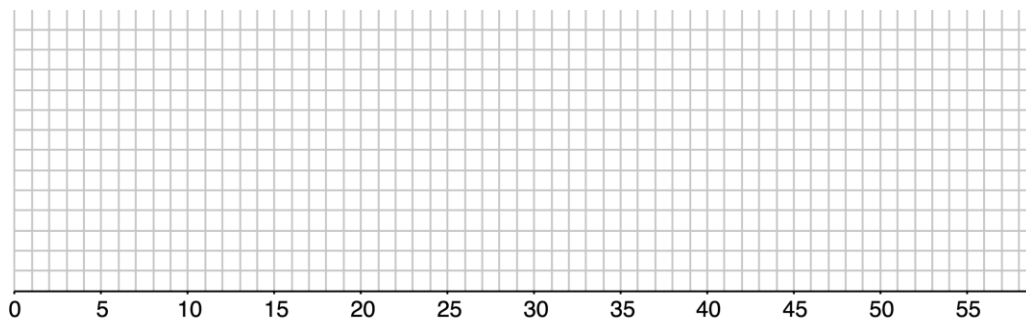
- Compare your box plots with your partner. Make any revisions if necessary.

### 10.1.5 Try It: Creating a Box Plot

1. The ages of the 21 restaurant customers who attended a special event at a local restaurant are shown.

27, 28, 39, 18, 18, 21, 32, 25, 46, 40, 40,  
22, 23, 24, 28, 30, 33, 35, 38, 42, 57

- a. Determine the five-number summary for the data set.
- b. Create a box plot using the data and five-number summary for the 21 restaurant customers.



- c. Does the length of each part of the box plot need to be the same?
- d. Why is the actual length of each part not equal, even though they each represent 25% of the data?
- e. What is the interquartile range for the data?
- f. What is the range?

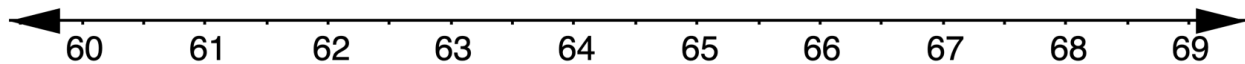
**10.1.6 Your Turn!**

1. Tom Brady is considered by most to be one of the best NFL quarterbacks of all time. The five-number summary for his pass-completion percentages from 2000 to 2009 is shown.

| Minimum | Maximum | $Q_1$ | $Q_3$ | Median |
|---------|---------|-------|-------|--------|
| 60.2    | 68.9    | 61.8  | 65.7  | 63.3   |

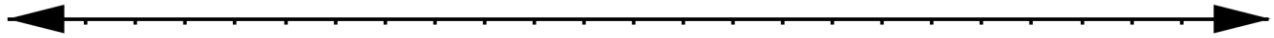
- a. Discuss with your partner why the data used to determine the five-number summary for Tom Brady's pass completion is more truthful than the data for golden retriever puppies, or the ages of the restaurant customers who attended a special event.
- b. Use the five-number summary to draw and label a box plot of the data set.

**Tom Brady's Completion Percentages 2000-2009**



- c. Determine an alternative scale that could be used to make the box plot.
- d. What scale would you use if the data values ranged from 10.3 to 48.4?

- e. Label the number line with your scale from Part D. Then, compare it with your partner.

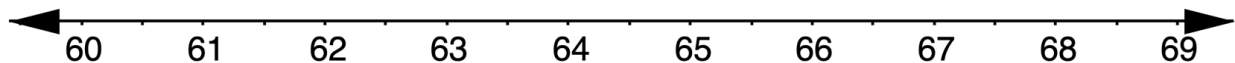


2. Tom Brady's pass-completion percentages from 2009 to 2019 are listed in the table.

| Year | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 |
|------|------|------|------|------|------|------|
| %    | 65.7 | 65.9 | 65.6 | 63.0 | 60.5 | 64.1 |
| Year | 2015 | 2016 | 2017 | 2018 | 2019 |      |
| %    | 64.4 | 67.4 | 66.3 | 65.8 | 60.8 |      |

- a. Create a box plot of Tom Brady's pass-completion percentages.

**Tom Brady's Pass-Completion Percentages 2009-2019**



- b. Determine the interquartile range for Tom Brady's pass-completion percentages.
- c. What is the range of pass-completion percentages?

### 10.1.7 Wrap-Up: Reflection

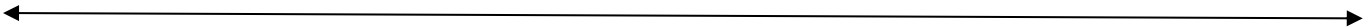
Place an **X** on the line to indicate your understanding of creating box plots. Then provide a brief explanation or evidence to support your self-assessment.

1. I can find the five-number summary of a data set.

I need help.  
I can't get started.

I'm getting there.

I understand  
and have evidence.



2. I can create an axis with the appropriate scale.

I need help.  
I can't get started.

I'm getting there.

I understand  
and have evidence.



3. I can find and interpret the interquartile range.

I need help.  
I can't get started.

I'm getting there.

I understand  
and have evidence.

